

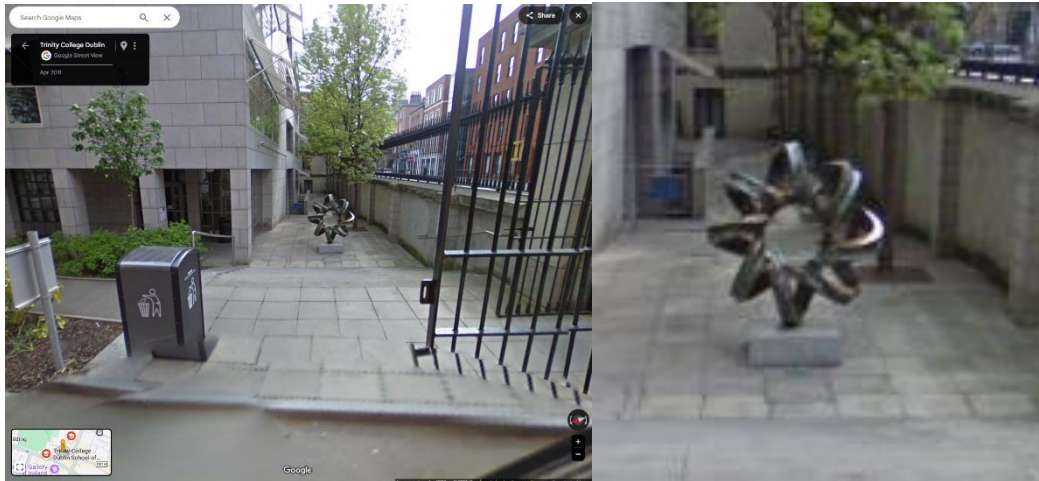
Augmented Reality

Assignment 1 - Report

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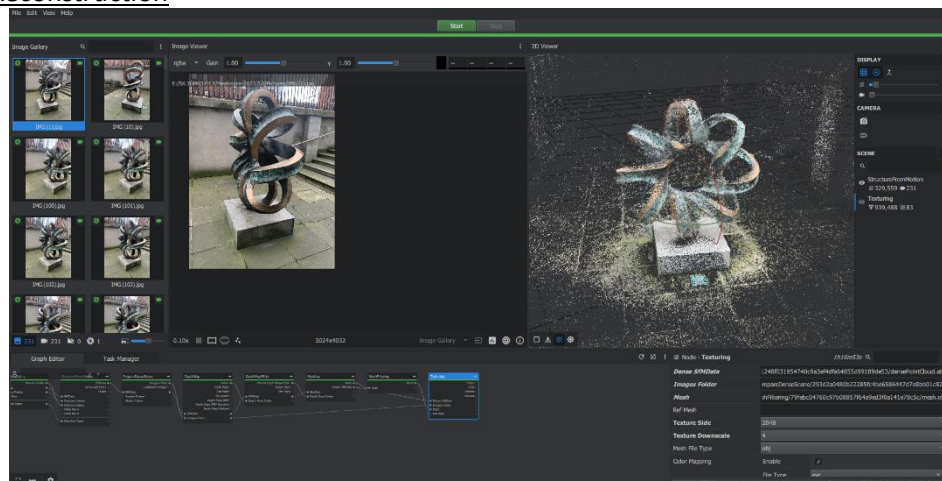
The objective of this assignment was to recreate a Dublin landmark as a detailed 3D model via Meshroom, import it into Unity to use in creating an AR scene, and then upload the model to Sketchfab.

I decided to create a 3D reconstruction of the Double Helix monument, located next to the Smurfit Institute of Genetics on the Trinity College campus.

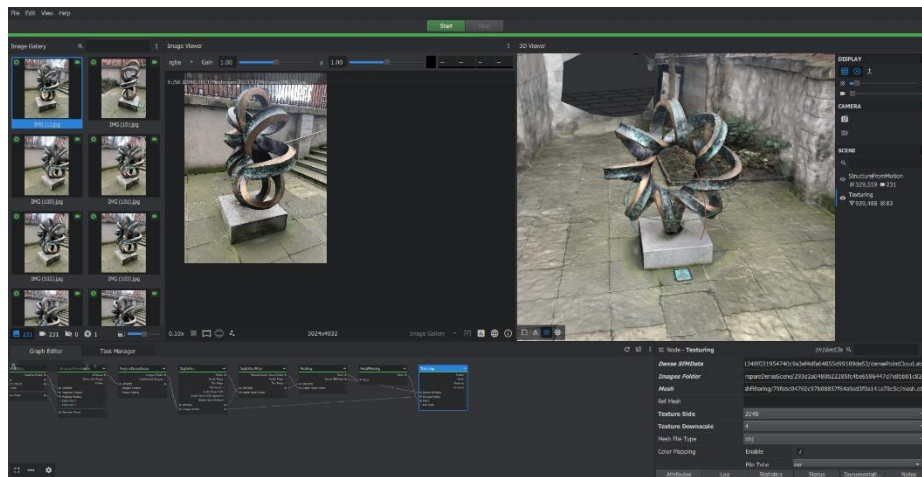


The image dataset contains 231 photos of the monument taken from different angles. The images had a 3024×4032 resolution and were captured using an iPhone Xs Max. As Meshroom couldn't process the images in their original *HEIC* format, they were converted to *JPG* using an online converter. The scan settings were set in accordance with the assignment guidelines. The computation process took approximately 3-4 hours, with the texturing step taking up the most time. From the process's completion, the following could be loaded and observed:

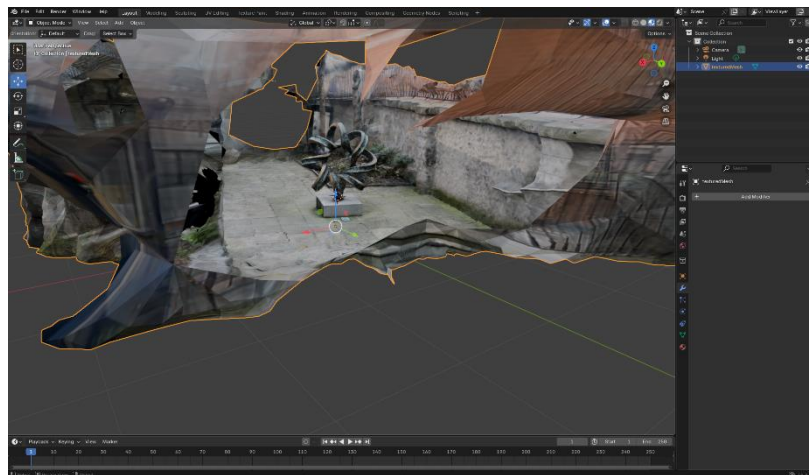
A Sparse Reconstruction



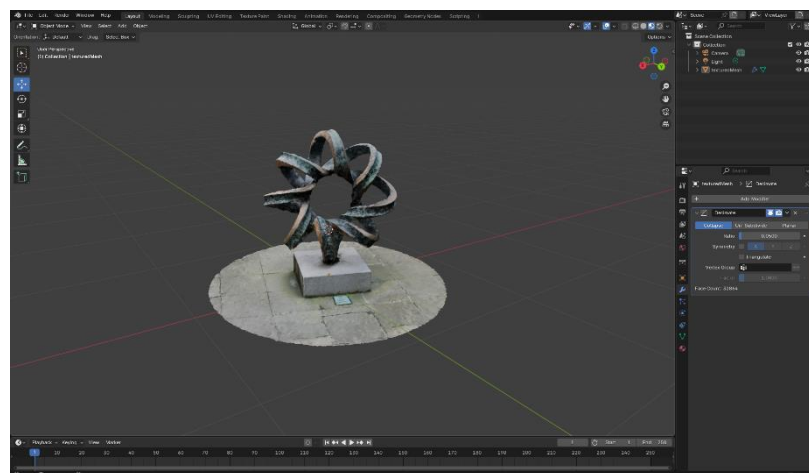
A Dense Reconstruction (textures included)



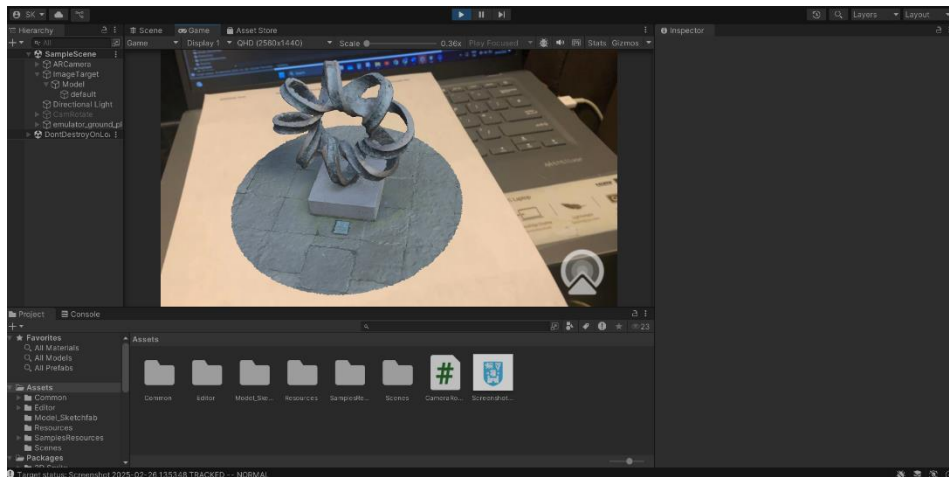
When importing the model as is to Blender, it is apparent that most of the surrounding area had also been modelled and textured.



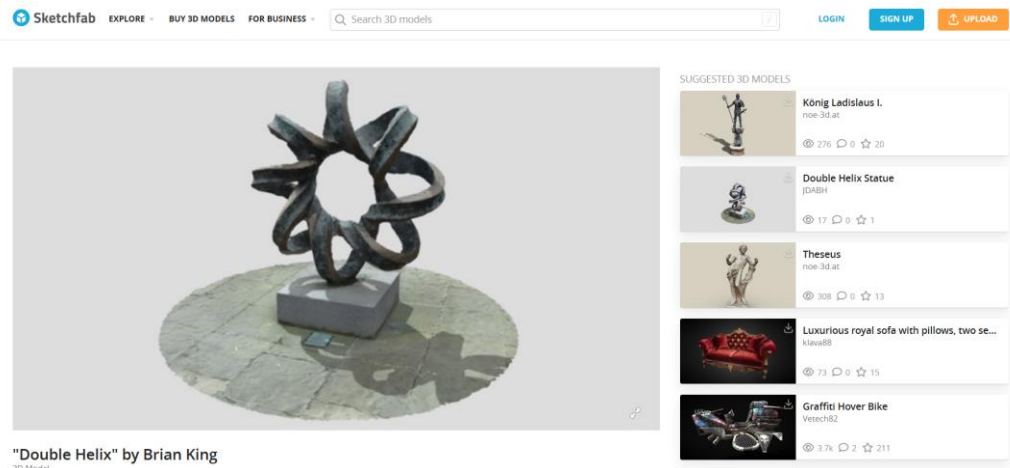
To get the monument by itself the model was cleaned up as instructed in the assignment guidelines.



From there, the model was utilised in an AR Unity scene-



-and subsequently uploaded to Sketchfab.



<https://sketchfab.com/3d-models/double-helix-by-brian-king-24820ba4b81d420091979085c8d46cae>

In the event of reattempting this assignment at a later date, I would aim to create a higher-quality model and utilise more powerful hardware to decrease computation time.